

Gemini Near Infra-Red Imager



Near IR Imager (NIRI)—The Engineering User Interface

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This document describes the operation of the engineering user interface for NIRI.

1 Introduction

The Engineering User Interface will be delivered with NIRI to Gemini. Its purposes are:

- to provide a user interface for testing and debugging NIRI;
- to provide a user interface before the standard NIRI user interface is available;
- to provide sample code which the designers of the NIRI standard user interface may inspect and reuse;
- to provide a user interface for the NIRI acceptance test procedure;
- to provide an interface to NIRI functions which the designers of the NIRI standard user interface may choose not to make available to the general users.

It is extremely important to read the warnings in Section 2.1, “Warnings (Very Important!),” on page 2!

1.1 Purpose

This document describes the operation of the NIRI engineering user interface.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the Gemini Observatory Control System (OCS) and the NIRI standard user interface;
- the developers of the GNIRS OIWFS components controller and other groups which will be using the design of the NIRI OIWFS
- the scientists, engineers, and technicians who will be designing and testing NIRI;

- the committee which will be reviewing the NIRI acceptance test procedure and who will be verifying that NIRI passes its acceptance tests;
- the early users of NIRI.

1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Warnings and General Information

2.1 Warnings (Very Important!)

There are a number of screens which contain functions which may damage NIRI or which may leave it in a state from which no software recovery is possible. These screens are easily distinguished from the standard NIRI screens, because they have a bright yellow title bar on a black background. The functions on these screens are intended for use by the qualified personnel only, and are not intended for general use. They include several functions which are capable of bypassing various safety features.

2.2 A Few Notes About Edd/Dm Interfaces

3 General User Commands

3.1 Starting the User Interface

3.2 The Introduction Screen

3.3 The Components Controller Main Screen

3.4 The On-Instrument Wavefront Sensor Main Screen

4 Restarting After the IOC Has Been Rebooted

4.1 Wait for initialization to complete

4.2 Restart screens

4.3 Look for RED

4.3.1 Set the Temperature Interlock

4.3.2 Start the Closed Cycle Coolers (if needed)

4.4 Look for YELLOW

4.4.1 Datum Mechanisms

5 Advanced User commands

The commands which are described in this section can damage the instrument or leave it in an unusable state. Use with extreme caution. *It is extremely important to read the warnings in Section 2.1, “Warnings (Very Important!),” on page 2!*

Notice: The screens which are described in this section do not meet the Gemini interface standards. They do not interact with the EPICS database using the mandatory CAD/CAR/APPLY/SIR interface. This is explicitly permitted in [33].

5.1 Starting the Low-Level Engineering Interface



5.2 The Main Engineering Screens

FIGURE 1. The Top Level CC Screen

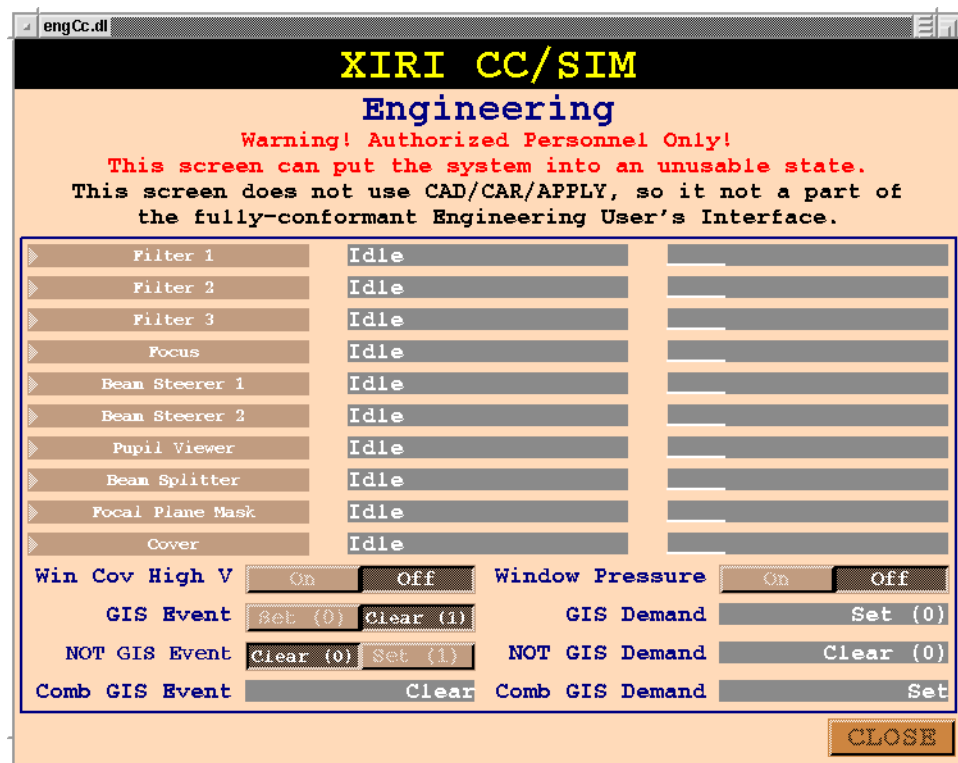
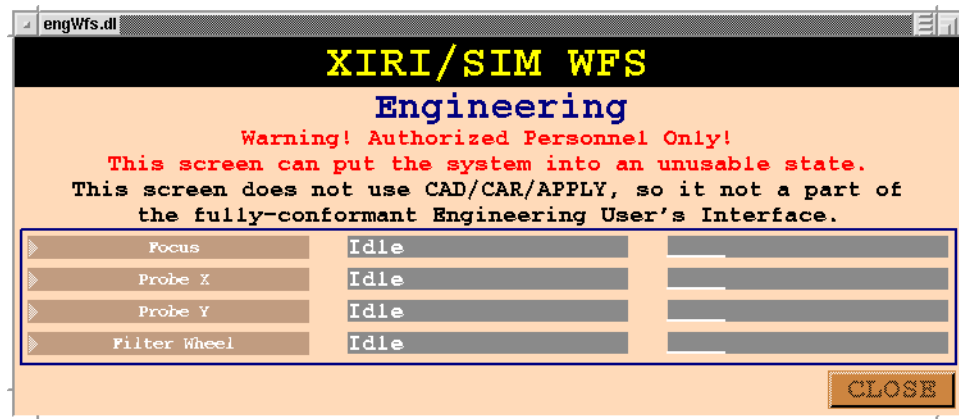


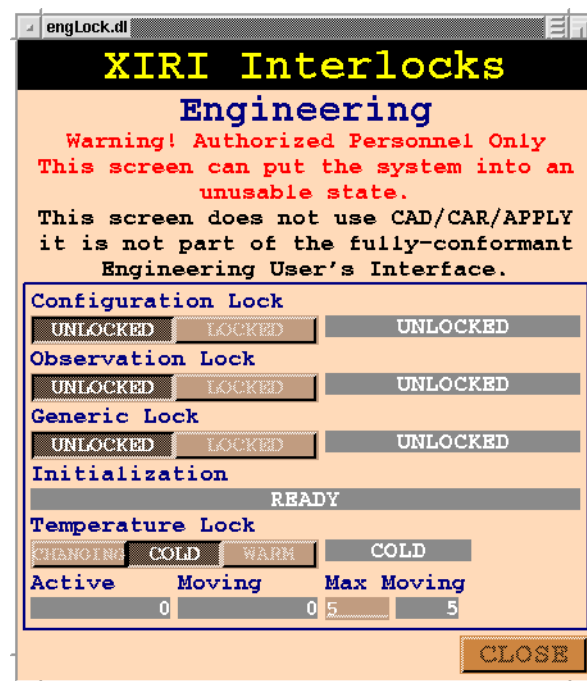
FIGURE 2. The Top Level OIWFS Screen



5.3 The Global Parameters Screen



5.4 The Engineering Interlocks Screen



5.5 The Low Level Mechanism Control Screen

engMech.dl

XIRI CC/SIM Filter 1

Engineering

Warning! Authorized Personnel Only!
 This screen can put the system into an unusable state.
 This screen does not use CAD/CAR/APPLY so it is not a
 part of the fully-conformant Engineering User's
 Interface.

Status Idle Error/Info 0			Sensors 1P (V) 2P (V) 3.997 0.000  -5.000 5.000 -5.000 5.000 Disable Disable Enable Enable	
Motor User Raw Stop New Pos. Cur Pos Pos Err 0 0 1 Low High Sens Tol 0 288000 0.300 Speed Acc T(s) Datumed 3000.0 1.0 0 Datum Redatum Datum/Diag Cycle End Cycle Diagnose Set ctr SW Count HW Count Use Set 0.0 0.0			1B (V) 2B (V) 3.998 0.000  -5.000 5.000 -5.000 5.000 Disable Disable Enable Enable	
Select home home 0.000 Ex Md Ps Ex Sen P Ex Sen B 0 0.000 0.000 Mod Pos Rd Sen P Rd Sen B 0 3.997 3.998 Update			Stripchart Related Screens External Tools CLOSE	

5.6 The Parameter Screen

engParams.dll

XIRI CC/SIM Filter 1

Engineering Parameters

Warning! Authorized Personnel Only!
This screen can put the system into an unusable state.

This screen does not use CAD/CAR/APPLY, so it is not a part of the fully-conformant Engineering User's Interface.

Cycle Parameters

	Toler	Set Exp Value				
	0.300	NO	YES			
Pos.	Exp1	Val1	Exp2	Val2		
A	13400	-0.779	0.000	0.000	0.000	P
		-0.779	0.000	0.000	0.000	B
B	-10600	-3.119	0.000	0.000	0.000	P
		-2.339	0.000	0.000	0.000	B
C	0	3.997	0.000	0.000	0.000	P
		3.998	0.000	0.000	0.000	B

Motor Control

Auto Off	Mot Pwr	Module	Reset Module
No	Yes	Auto On	Off
		Good	Reset

Motion / Homing Parameters

Backlash	DSLP	DSLB	THRP	WTHP
1000	0	0	0.000	400
Timeout	DOFP	DOFB	THRB	WTHB
120	0	0	2.000	400
DOF1	LOPR	HOPR	ZERO	
-30	0	288000	0	

Reload LUTs:

5.7 The Temperature Control Screen

engTmp.dll

XIRI/SIM Temperature control Engineering

Warning! Authorized Personnel Only!
 This screen can put the system into an unusable state.
 Center the science focusing stage before cooling or warming.
 This screen does not use CAD/CAR/APPLY, so it not a part of
 the fully-conformant Engineering User's Interface.

Temperature Sensors		Temperature Controllers			
		OIWFS	Img Buf Mass	Cold Pl Center	
Serial I/O		Serial I/O			
Top Strap	300.00	Temperature	300.00	300.00	300.00
Bottom Strap	300.00	Heater %	0.00	0.00	0.00
Cryopump	300.00	Set Point	0.00	0.00	0.00
Cold Pl Edge	300.00	Gain	0.00	0.00	0.00
CC Cabinet	300.00	Rate	0.00	0.00	0.00
WFS Cabinet	300.00	Reset	0.00	0.00	0.00
7	300.00	Range	Off	Off	Off
8	300.00	Tune	Manual	Manual	Manual
Units	K	Units	K	K	K

Closed Cycle Coolers					
Cooling	Off	Low	High	State	Off
Low	30000			Pos	0
High	55000			Vel	1000
Enable 1	Auto	On	Off	Jog	0
Enable 2	Auto	On	Off	Dmov	1
				Abort	
Timer	Cancel			Heater Mode	Normal

CLOSE

